

# ShopfloorGPT Deployment using Kubernetes, Terraform, GitLab

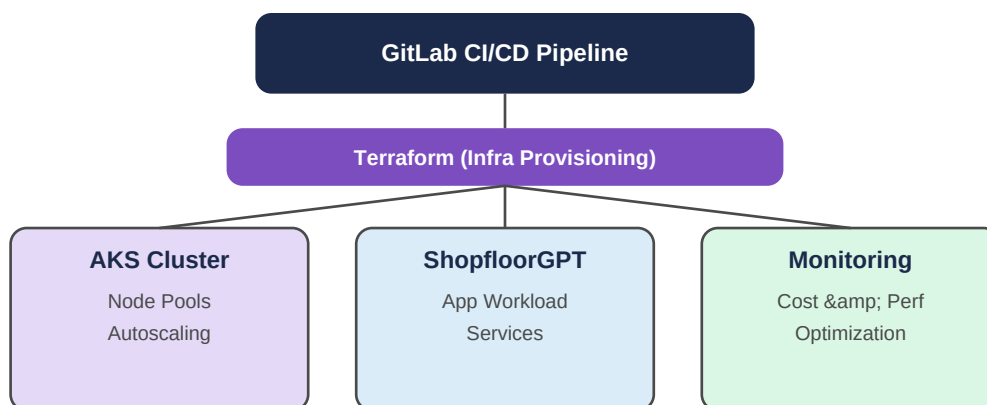
## DevOps & Cloud Engineering Case Study

|                                       |                                          |                                      |
|---------------------------------------|------------------------------------------|--------------------------------------|
| <b>70%</b><br>Faster Infra Setup Time | <b>80%</b><br>Less Manual Release Effort | <b>30%</b><br>Lower AKS Compute Cost |
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### Overview

Deployed ShopfloorGPT on Azure Kubernetes Service (AKS) using Terraform for infrastructure provisioning and GitLab CI/CD for automated builds and releases — turning a manual, ad-hoc deployment process into a fully automated, reproducible pipeline.

### Architecture



### Key Deliverables

- Deployed ShopfloorGPT on AKS using Terraform, making infrastructure fully reproducible and reducing environment setup time by 70%.
- Implemented GitLab CI/CD pipelines for automated builds and deployments, cutting manual release effort by 80% and enabling consistent, repeatable releases.
- Optimized and maintained the AKS cluster on an ongoing basis, improving workload performance while lowering compute costs by 30%.
- Standardized infrastructure and release workflows into version-controlled Terraform and GitLab pipeline definitions, removing manual, error-prone deployment steps.

### Outcome

A fully automated, cost-efficient deployment pipeline for ShopfloorGPT — from infrastructure provisioning through to production release — with reproducible environments, faster releases, and a leaner, better-optimized AKS footprint.

### Skills & Tools

Kubernetes (AKS) · Terraform · GitLab CI/CD · Azure · Infrastructure as Code (IaC) · Cloud Cost Optimization · Cluster Management · DevOps Automation